Java Programming

WEEK 3 - Data Structures & File Handling

Focus: Learn data management and algorithmic thinking with collections.

Topics Covered:

- Java Collections Framework (List, Set, Map)
- ArrayList, HashMap, and HashSet
- Sorting and Searching algorithms
- File Handling (Read/Write files using `FileReader` / `FileWriter`)
- Exception Handling (try, catch, finally, throw, throws)

Assignments:

1. **Student Marks Record Program**
    - Store data in an `ArrayList` and calculate highest & lowest marks.
2. **File I/O Assignment**
    - Read from a text file, count word frequency, and write output to another file.

Mini Project:

Employee Management System (Using Collections + File I/O)

- Store employee details (ID, Name, Department, Salary)
- Save and load data from `.txt` file
- Use exception handling for invalid inputs

Suggested Video: Java Collections Framework Tutorial